Red Hat Summit

Multi-site disaster recovery for healthcare systems with InfoScale for Kubernetes and Red Hat OpenShift Virtualization

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Disaster Recovery for Virtual Machines on OpenShift Virtualization

Organizations are increasingly running virtual machines on Kubernetes platforms such as OpenShift Virtualization.

Why this matters

- Protection of stateful VM workloads
- Data replication across clusters
- Application availability during infrastructure failures

In this lab you will learn

- How to implement Disaster Recovery for VM workloads
- How to configure replication between OpenShift clusters
- How to execute a failover scenario

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Challenges with Traditional DR Approaches

Traditional Disaster Recovery solutions were designed for traditional virtualization environments.

While effective in traditional environments, these approaches present challenges in modern Kubernetes platforms.

Key challenges

- Storage vendor lock-in
- DR tools not designed for Kubernetes workloads
- Complex infrastructure-level failover procedures
- Limited flexibility across heterogeneous storage systems
- Infrastructure-level failover complexity
- Lack of workload-aware recovery orchestration

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Lab Scenario

A healthcare provider runs a patient management system inside a virtual machine on OpenShift Virtualization.

This application stores critical patient data and must remain available even if infrastructure failures occur.

Infrastructure environment

- Primary OpenShift cluster running the application
- Secondary OpenShift cluster for Disaster Recovery
- Persistent VM storage provided by InfoScale for Kubernetes

Lab objective

- Implement Disaster Recovery for the VM workload
- Validate recovery by performing a failover to the secondary cluster

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How InfoScale for Kubernetes Solves This

InfoScale for Kubernetes provides a software-defined storage and disaster recovery layer for Kubernetes and OpenShift Virtualization environments.

Key advantages

- Software Defined Storage for Kubernetes workloads
- Replication across clusters for Disaster Recovery
- Consistent storage layer across heterogeneous storage arrays
- Preserves existing SAN infrastructure investments during VMware-to-OpenShift (V-to-O) migrations.
- Orchestrated failover workflows for VM workloads
- Minimum RPO and RTO

What this enables

- Protection of virtual machine persistent storage
- DR operations integrated with Kubernetes environments
- Recovery of applications across OpenShift clusters
- Simplified DR operations for OpenShift Virtualization
- Flexibility across different storage vendors
- Reliable recovery of stateful VM workloads

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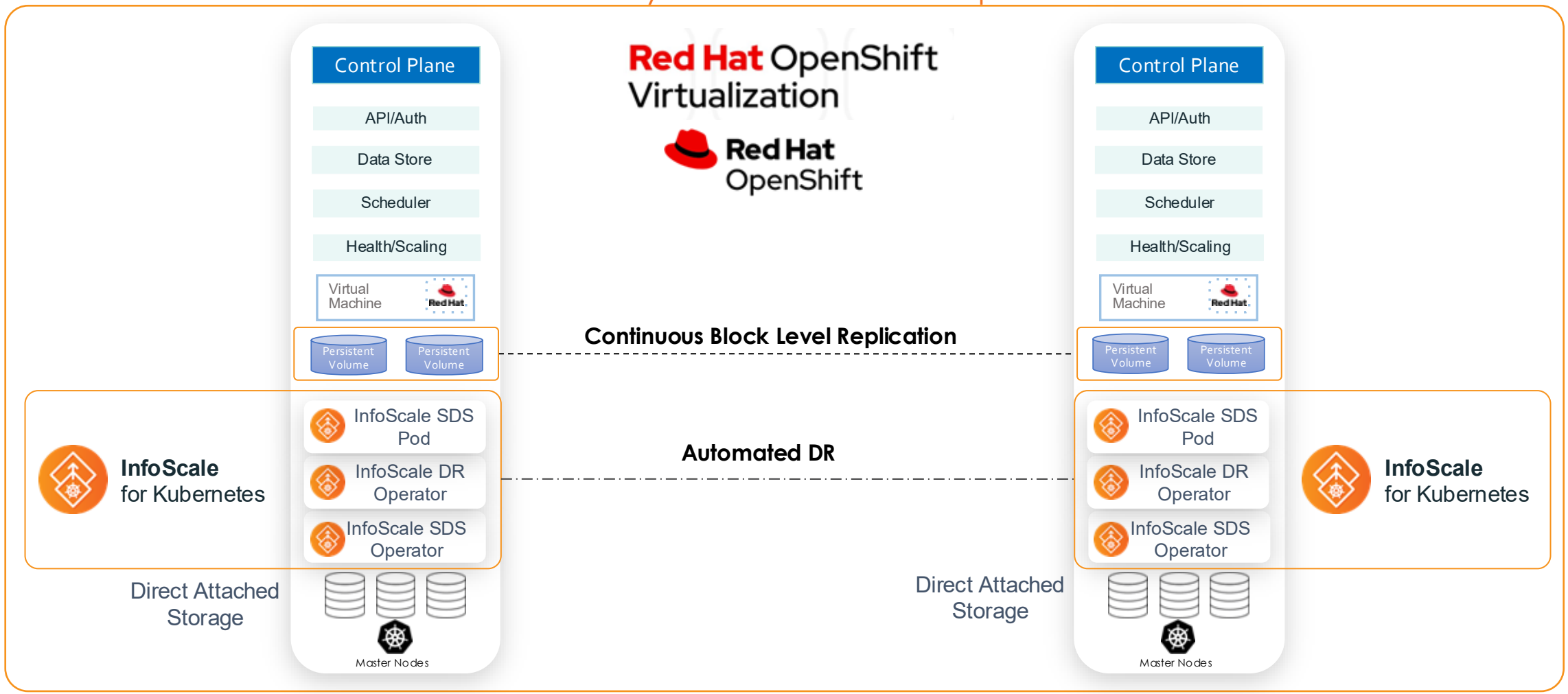
Lab Architecture

This lab environment simulates a Disaster Recovery scenario between two OpenShift clusters.



Lab Workflow

In this hands-on lab we will walk through the process of implementing Disaster Recovery for a virtual machine workload running on OpenShift Virtualization.

Lab steps

- Verify the InfoScale for Kubernetes installation across both clusters
- Install and configure the InfoScale DR Manager and establish inter-cluster communication
- Validate a stateful healthcare VM application running on OpenShift Virtualization
- Configure synchronous data replication between the primary and secondary clusters
- Create and execute a disaster recovery plan using InfoScale CRDs
- Validate application availability and data consistency after a DR failover

Goal

- Demonstrate end-to-end Disaster Recovery for virtual machines running on OpenShift Virtualization.

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Key Takeaways and Next Steps

Key Benefits Demonstrated:

- Achieve near-zero RPO and RTO for OpenShift Virtualization.
- Simplify complex DR operations using Kubernetes native tools.
- Leverage existing SAN storage investments with InfoScale SDS.

Call to Action:

- Visit our InfoScale Booth for deeper technical discussions and demos.
- Scan the QR code to access the lab materials and documentation.

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Thank you & Q & A

Connect with the Speakers

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